

IE University
Master in Business Analytics & Data Science

MASTER'S THESIS

Mean-Variance Dynamics and Market Efficiency in Silver

A Multi-Model Forecasting Framework

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Abstract

This thesis asks how much of silver’s short-term price behaviour can be forecast, and frames the question through the efficient-market hypothesis (EMH). Silver is modelled at the weekly (Friday-to-Friday) horizon over 2015–2026, and the analysis is factorised into the two moments: the conditional *mean* of returns and the conditional *variance*—efficiency constrains the former but not the latter. For the mean, a deliberate ladder of models of increasing flexibility—ARIMA/ARIMAX, VAR, mixed-data-sampling (MIDAS) regressions, random forests, gradient boosting, and an LSTM—is tested against a common random-walk-with-drift benchmark, expanding the information set from silver’s own past (weak-form EMH) to a broad set of public information (semi-strong-form EMH): cross-asset returns, silver futures positioning, macroeconomic releases, technical indicators, and sentiment from Reddit and financial news. For the variance, weekly realised volatility is modelled with GARCH and the heterogeneous autoregressive (HAR) model, and the question of interest is whether sentiment improves the forecast. The central finding is a single coherent statement: no model beats the drift benchmark for the conditional mean by a significant margin—weak- and semi-strong-form efficiency hold for weekly silver returns—yet the conditional variance is strongly and significantly forecastable, with sentiment adding a small but robust increment. Predictable volatility alongside an unpredictable mean is exactly what the EMH permits, so the two results are complementary rather than contradictory.

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1 Introduction

Silver lives a double life. It is an industrial input—roughly half of annual demand goes into photovoltaics, electronics, brazing alloys, and medical uses, tying its price to the global manufacturing cycle—and, at the same time, a monetary metal held as a store of value, an inflation hedge, and a more volatile cousin of gold. On top of both sits a retail following that can sometimes turn silver into a meme trade: in late January 2021, Reddit’s `r/WallStreetSilver` drove spot from roughly \$25 to \$30 an ounce within days before it faded. This blend of industrial demand, monetary hedging, and episodic retail attention makes silver an unusually rich laboratory for the oldest question in empirical finance: how much of a price can be forecast?

That question has a canonical answer, and it is mostly negative. The efficient-market hypothesis (Fama, 1970, 1991) holds that prices already embed available information, so successive price changes are approximately unpredictable. It is conventionally stated in three nested forms, graded by the information set a forecaster is allowed to use:

- **Weak form** — past prices and returns carry no exploitable signal.
- **Semi-strong form** — no *public* information does either: not prices, fundamentals, news, or macroeconomic data.
- **Strong form** — not even *private*, inside information does.

Each form nests the one before it, so the claims grow progressively stronger. For a liquid, globally traded metal, weak- and semi-strong-form efficiency are the natural hypotheses to take to the data, and they are the two this thesis tests.

Empirically, silver’s *price* behaves like a random walk with drift: it is non-stationary, and shocks accumulate rather than mean-revert (Figure 1). Returns, however, are stationary—they fluctuate around a stable long-run mean rather than wandering without bound. The object worth modelling is therefore not the price level but its increment, the weekly log-return $r_t = \log P_t - \log P_{t-1}$; stationarity is what makes the forecasting problem well-posed, giving a fixed quantity to predict and a natural yardstick—that same prevailing average—to judge any forecast against.

The weekly (Friday-to-Friday) horizon is a deliberate choice of frequency, made where the information aligns. Futures-market positioning is reported weekly and macroeconomic releases monthly, while daily sentiment is too sparse and noisy to inform a forecast and daily returns are dominated by microstructure noise; a weekly horizon is coarse enough to average out that noise yet fine enough to leave roughly 600 observations over 2015–2026—enough to train and properly out-of-sample-test the machine-learning models. Weekly realised volatility, the target of the second-moment analysis, is in turn

aggregated from daily returns.

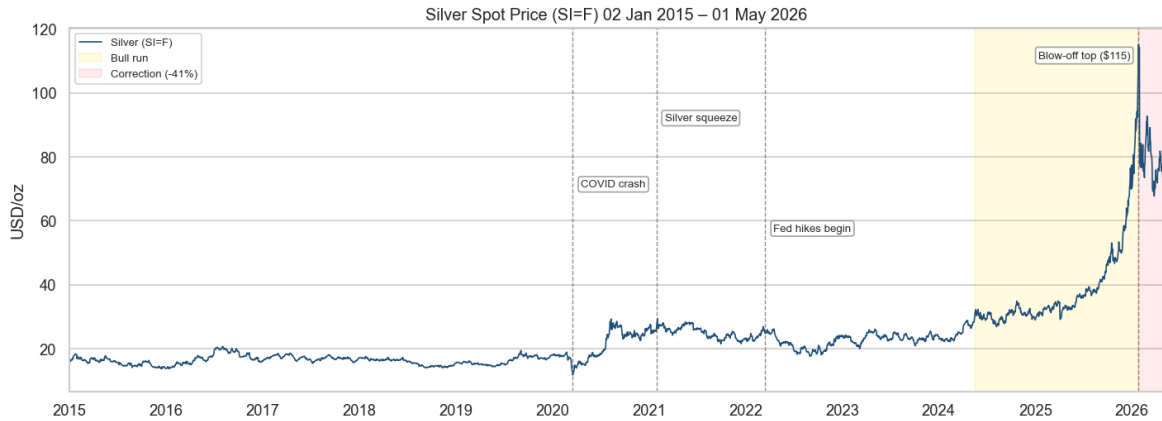


Figure 1: Silver front-month futures (SI=F), daily, January 2015–May 2026. The annotated episodes punctuate the sample: the COVID-19 crash (March 2020), the r/WallStreetSilver squeeze (January 2021), the start of the Federal Reserve’s hiking cycle (March 2022), and the 2024–2026 bull run that carried spot to an all-time high near \$115/oz before a sharp (−41%) correction. The level is non-stationary and shocks accumulate; the thesis models its stationary weekly log-return rather than the level itself.

The efficiency question then becomes whether r_t can be forecast—and silver’s own past offers little to go on. Weekly returns carry almost no linear autocorrelation, so a linear own-history model such as ARIMA has little to grip; a formal model search duly collapses it to a constant mean. Yet the returns are not independent noise either, because their *magnitude* is persistent—volatile weeks cluster together—so structure remains, only in the variance rather than the mean. The thesis follows this thread: it asks first whether silver’s own history forecasts the return, then whether the broad public-information set a real forecaster would possess does—cross-asset returns (gold, the US dollar, copper, equities, oil, and the VIX), futures-market positioning, macroeconomic releases, technical indicators, and retail and news sentiment—and, separately, whether the predictable variance can be modelled.

Research questions. The thesis is built around four questions:

- **Q1 (weak form).** Are weekly silver returns forecastable from their own past?
- **Q2 (semi-strong form).** Does expanding the information set to public covariates help—and does it matter whether the relationship is assumed linear or allowed to be nonlinear?
- **Q3 (the second moment).** Is weekly realised volatility forecastable, and if so by how much?
- **Q4 (sentiment).** Does retail and news sentiment add incremental predictive

content to either the conditional mean or the conditional variance?

Approach. To answer these questions, the analysis estimates a deliberate ladder of models of increasing flexibility against a common benchmark and a unified evaluation protocol. The benchmark throughout is the prevailing-mean “drift” forecast—the random-walk-with-drift, equivalent to ARIMA(0,0,0)—which is the correct null for a return target (Campbell & Thompson, 2008; Welch & Goyal, 2008). For the conditional mean, the ladder runs from classical time-series models (ARIMA/ARIMAX, VAR, and MIDAS regressions) through tree ensembles (random forests and gradient boosting) to a recurrent neural network (LSTM); for the conditional variance it runs GARCH and the heterogeneous autoregressive (HAR) model of Corsi (2009) alongside the same tree ensembles.

2 Conceptual Framework

This section fixes the notation used throughout and states, precisely, the null hypothesis that each model class is built to test. The aim is to keep the language exact without overclaiming: the empirical results are framed as tests of the conditional mean and the conditional variance of weekly silver returns, not as claims about the full return distribution.

2.1 Prices, returns, and the random walk

Let P_t be the silver price and $p_t = \log P_t$ the log price. The weekly log-return is the increment in the log price,

$$r_{t+1} = p_{t+1} - p_t, \quad \text{equivalently} \quad p_{t+1} = p_t + r_{t+1}, \quad (1)$$

so the price level and the return are two views of the same process: the level accumulates shocks, while returns *are* the shocks. A random walk with drift writes the level as $p_{t+1} = p_t + \mu + \varepsilon_{t+1}$ with $\mathbb{E}[\varepsilon_{t+1}] = 0$. Silver log prices are best described as *random-walk-like*—non-stationary, with shocks that accumulate—which is the reason the modelling target throughout is the stationary increment r_t rather than the level. The safer wording avoids the *strict* random walk: the innovations are not independent and identically distributed, because their variance clusters (Section 2.4).

2.2 The conditional mean and the information set

Let F_t denote the information available at time t . For the weak-form test, this is silver’s own past, $F_t = \{r_t, r_{t-1}, r_{t-2}, \dots\}$. Returns satisfy the *martingale difference* property

with respect to F_t if their conditional mean is constant,

$$\mathbb{E}[r_{t+1} \mid F_t] = \mu, \quad (2)$$

or, equivalently, the demeaned return $r_{t+1} - \mu$ is a *martingale difference sequence* (MDS),

$$\mathbb{E}[r_{t+1} - \mu \mid F_t] = 0. \quad (3)$$

Past returns do not move the expected next-week return away from its long-run average. This is a statement about the first moment *only*; it does not assert that returns are i.i.d. The semi-strong test expands the information set to public covariates,

$$G_t = \{F_t, \text{ gold, USD, copper, VIX, oil, COT, macro, technicals, sentiment}\}, \quad (4)$$

and asks whether the martingale-difference property survives the larger conditioning set,

$$\mathbb{E}[r_{t+1} \mid G_t] = \mu. \quad (5)$$

ARIMA tests whether silver returns are a martingale difference with respect to their own past, Equation (2); ARIMAX and the other models test whether that property survives once the information set is expanded to public covariates, Equation (5). Operationally, these nulls are tested as “does the model beat the prevailing-mean (drift) benchmark out of sample?”—a forecast that beats the drift would contradict the corresponding null.

2.3 Linear and nonlinear alternatives

The alternative to Equation (5) can be specified narrowly or broadly. A linear model entertains an additive alternative,

$$\mathbb{E}[r_{t+1} \mid G_t] = \mu + \beta' G_t, \quad (6)$$

so ARIMAX, VAR, and MIDAS ask whether public covariates enter the conditional mean linearly and additively. Nonlinear models entertain the broader alternative

$$\mathbb{E}[r_{t+1} \mid G_t] = f(G_t), \quad (7)$$

in which the conditional mean can respond to thresholds and interactions among the covariates rather than to each one additively—a signal might be invisible in any single predictor yet surface in a combination of them. Random forests, gradient boosting, and the LSTM search this larger space. The logic of the ladder is therefore cumulative: if even flexible nonlinear models fail to beat the drift benchmark out of sample, the

martingale-difference interpretation is *strengthened* rather than merely assumed from linear tests. A complementary own-history check, the generalised spectral test (Hong & Lee, 2003), probes for nonlinear departures from Equation (2) that linear autocorrelation diagnostics cannot see.

2.4 The conditional variance

Efficiency constrains Equation (5); it places no restriction on the conditional variance $\text{Var}(r_{t+1} | F_t)$. The two can diverge: the mean may be (approximately) constant while the variance moves through time, which is exactly what volatility clustering means—a large move today says little about the sign or size of next week’s return but does signal that next week is likely to be more volatile. Strict white noise would require independence in r_t , r_t^2 , and $|r_t|$ alike; the MDS property in Equation (2) restricts only the first of these. Weekly silver returns are close to white noise in the mean but not strict white noise, because squared returns are strongly autocorrelated. The predictable structure is therefore concentrated in the second moment, and the squared return is its natural proxy. By definition it splits into the variance and the squared mean, and at the weekly horizon the latter is negligible:

$$\mathbb{E}[r_t^2] = \underbrace{\text{Var}(r_t)}_{\approx 1.9 \times 10^{-3}} + \underbrace{(\mathbb{E}[r_t])^2}_{\approx 7 \times 10^{-6}} \approx \text{Var}(r_t). \quad (8)$$

The squared mean is only 0.4% of the variance, so a squared return is—to that approximation—a model-free measure of variance. This is what the volatility chapter (Section 7) builds on, summing daily squared returns into weekly realised volatility. Figure 2 shows both moments at once for the weekly silver return.

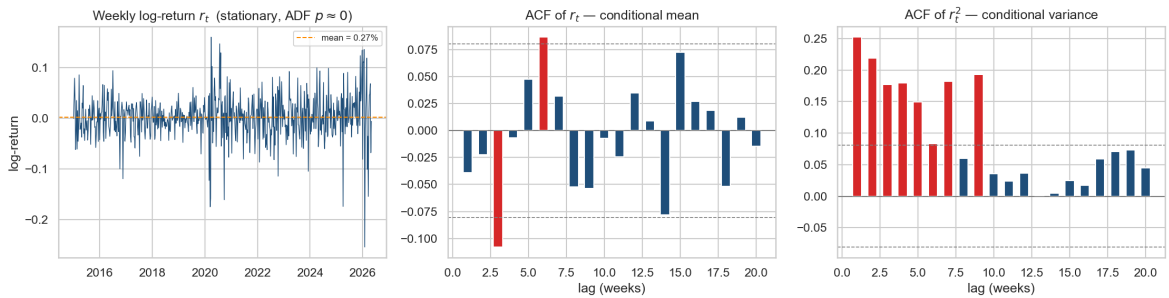


Figure 2: The two moments of the weekly silver log-return, 2015–2026. *Left*: the return series—stationary, fluctuating around a small positive drift (mean $\approx 0.27\%$ per week). *Centre*: the sample autocorrelation of returns lies almost entirely within the 95% white-noise band (dashed), with only isolated lags breaching it—consistent with the martingale-difference null of Equation (2), so a linear own-history model has little to exploit. *Right*: the autocorrelation of *squared* returns is large and positive across the first several lags—the volatility clustering of Section 2.4. The mean is close to unpredictable; the variance is not.

2.5 Direction versus magnitude

A final distinction governs how forecasts are scored. Directional accuracy asks whether $\text{sign}(\hat{r}_{t+1}) = \text{sign}(r_{t+1})$ more often than chance, which is *not* the same as beating the drift benchmark under squared-error loss: a model can call the sign correctly yet fail to improve out-of-sample R^2 . Because efficiency is a conditional-mean claim, the magnitude axis (RMSE, out-of-sample R^2 , and Diebold–Mariano against the drift) is load-bearing, while directional accuracy—and its significance test, Pesaran–Timmermann (Pesaran & Timmermann, 1992)—is a secondary, economically motivated lens. Directional gains are interesting but do not by themselves reject the martingale-difference null.

3 Literature Review

This thesis sits at the intersection of market efficiency, commodity forecasting, volatility modelling, and text-based investor attention. Its central distinction is between two kinds of predictability. The first is *conditional-mean* predictability: whether next week’s silver return can be forecast from past prices or public information. This is the economically load-bearing question for the efficient-market hypothesis. The second is *conditional-variance* predictability: whether the size of next week’s move can be forecast, even if its sign and expected value cannot. The literature suggests that these two forms of predictability should not be treated as contradictory. A market can be efficient in the mean while still displaying persistent volatility, because the EMH restricts expected returns, not time-varying risk.

3.1 Return predictability, market efficiency, and public information

The natural starting point is the martingale logic of Samuelson (1965) and the efficient-market taxonomy of Fama (1970, 1991). In an informationally efficient market, prices incorporate available information quickly enough that price changes are difficult to forecast from that same information. The weak form asks whether past prices and returns contain exploitable signal; the semi-strong form asks whether a broader public information set does. That the weak-form null can be rejected is itself well established—Lo and MacKinlay (1988), with a variance-ratio test on *weekly* returns, reject the random walk for US equity indices—so own-history predictability at the weekly horizon is possible in principle, and testing for it in silver is a genuine question rather than a foregone conclusion. For this thesis, the implication is direct: an ARIMA model tests whether silver’s own return history moves the conditional mean, while ARIMAX, VAR, MIDAS, machine-learning models, and the LSTM test whether public covariates do. If none improves materially on the prevailing-mean drift benchmark out of sample,

the result is evidence in favour of weak- and semi-strong-form efficiency at the weekly horizon.

Commodity markets complicate this simple null, because prices reflect both financial and physical forces. Deaton and Laroque (1992) document the extreme persistence and spike-like behaviour of commodity prices, which makes price levels poor forecasting targets and motivates modelling returns instead. At the same time, commodity futures returns have been linked to hedging pressure, inventories, basis, and risk premia (Bessembinder, 1992; G. Gorton & Rouwenhorst, 2006; G. B. Gorton et al., 2013; Szymanowska et al., 2014). These papers justify including futures positioning and cross-asset information, but they do not imply that a single weekly silver return should be easy to forecast out of sample. The distinction matters: variables may explain risk premia or cross-sectional commodity returns over longer horizons without generating a reliable one-week forecast for silver’s conditional mean.

Silver is also not simply “gold with more volatility.” Precious-metals research shows that gold, silver, platinum, and palladium respond differently to macroeconomic conditions. Batten et al. (2010), for example, find that the macroeconomic drivers of volatility differ across precious metals, while Zhang and Wei (2010) document important linkages between gold and oil. This supports the thesis’s broad public-information set—gold, the dollar, copper, oil, equities, the VIX, rates, inflation expectations, industrial production, and COT positioning—while also warning against assuming that silver’s behaviour can be inferred from gold alone. The CFTC’s Commitments of Traders data are especially relevant because they separate commercial and non-commercial positioning, making them a natural proxy for hedging pressure and speculative sentiment.

The benchmarking literature is equally important. Welch and Goyal (2008) show that many variables that appear to predict returns in sample fail to beat the historical average out of sample. Campbell and Thompson (2008) argue that small gains are possible when forecasts are disciplined by theory, but they retain the historical mean as the central benchmark and out-of-sample R^2 as the natural effect-size measure. This thesis follows that discipline. The drift forecast is not a weak straw man; it is the operational version of the EMH null for a return target.

The adaptive-markets hypothesis of Lo (2004) provides a useful interpretation for unstable or episodic findings. Under the AMH, market efficiency is not a fixed state but an evolutionary outcome shaped by competition, learning, institutional change, and changing trader populations. A temporary directional signal around a retail-attention episode, a crisis, or a commodity squeeze would therefore not be surprising. What matters is whether it survives out of sample, across regimes, and against a benchmark. This is why the thesis combines expanding and rolling windows, subperiod robustness

checks, and a separate interpretation of directional accuracy.

3.2 Mixed-frequency information and nonlinear forecasting models

The semi-strong test raises a practical problem: the information set is observed at different frequencies. Silver and cross-asset prices are daily, COT positioning is weekly, macroeconomic variables are monthly and released with delays, and sentiment can be daily but sparse. MIDAS regression was developed precisely for this kind of problem. Ghysels et al. (2004, 2007) show how high-frequency lags can enter a low-frequency forecast through a parsimonious lag-weight function rather than a large unrestricted lag set. Andreou et al. (2013) demonstrate the value of daily financial information for macroeconomic forecasting. Here, the logic is inverted: monthly macro releases and daily financial and sentiment variables are aligned to a weekly silver-return target. MIDAS is therefore not an exotic addition; it is the appropriate linear bridge between mixed-frequency public information and the weekly EMH test.

Machine-learning models extend the alternative hypothesis from linear to nonlinear predictability. Random forests and gradient boosting can detect thresholds and interactions; LSTMs can learn sequential patterns without hand-specifying every lag. This is important because a failure of ARIMAX or VAR alone would only reject simple linear predictability, not nonlinear predictability. The deeper forecasting literature, however, urges caution. Sezer et al. (2020) review a large body of deep-learning studies in financial time series and find many reported gains, but the broader forecasting literature shows that more complex methods do not automatically beat simpler benchmarks, and that rankings often depend on the evaluation metric and test design. The thesis therefore uses machine learning not as a promise of superior performance, but as a stricter test of efficiency: if flexible nonlinear models also fail to beat the drift, the mean-efficiency conclusion becomes stronger.

3.3 Directional accuracy versus magnitude accuracy

A separate strand of the literature evaluates whether forecasts get the *direction* of returns right, rather than their magnitude. Pesaran and Timmermann (1992) provide a formal test for market-timing ability, and directional metrics remain popular in economics and finance because traders often care about whether prices rise or fall. But directional accuracy is not the same object as RMSE, MAE, or out-of-sample R^2 . A model can call slightly more signs correctly while still making large errors on the weeks that matter most; conversely, a model can reduce squared error without producing a useful trading rule. Leitch and Tanner (1991) make the broader point that conventional statistical

error measures and economic profitability can rank forecasts differently. Christoffersen and Diebold (2006) similarly treat direction-of-change forecasting as a distinct problem from point forecasting.

For an EMH thesis, this distinction is crucial. The weak- and semi-strong-form hypotheses are fundamentally about whether the conditional expected return moves away from its benchmark value. Therefore RMSE, out-of-sample R^2 , and the Diebold–Mariano test against the drift are the primary evidence. Directional accuracy is still informative, especially under the adaptive-markets interpretation: a transient sign signal may reveal episodic market structure, behavioural pressure, or regime dependence. But unless that signal improves magnitude forecasts, survives robustness checks, and is economically exploitable after costs, it should be described as directional dependence, not as a rejection of mean efficiency.

3.4 Variance predictability: GARCH, realised volatility, and HAR

The volatility literature begins from the opposite empirical fact: although returns are hard to forecast, their magnitudes are persistent. The ARCH model of Engle (1982) and the GARCH extension of Bollerslev (1986) formalised volatility clustering by allowing conditional variance to depend on past shocks and past variance. This is now one of the most established facts in financial econometrics. For this thesis, the implication is that predictable volatility does not weaken the EMH result. It strengthens the mean-variance interpretation: silver may be efficient in expected return while still risky in a forecastable way.

Realised-volatility models sharpen the target by replacing latent volatility with a model-free proxy built from higher-frequency squared returns. Barndorff-Nielsen and Shephard (2002) and Andersen et al. (2003) helped establish realised variance as a practical object for volatility measurement and forecasting. The HAR model of Corsi (2009) then became a benchmark because it captures long-memory-like volatility persistence using simple averages over short, medium, and long horizons. Patton (2011) adds an important evaluation point: because realised volatility is itself a noisy proxy for latent variance, forecast comparisons should use losses that are robust to measurement error, especially QLIKE. This justifies the thesis’s use of GARCH and HAR as volatility benchmarks and QLIKE-based Diebold–Mariano tests as the main variance-comparison evidence.

3.5 Sentiment, attention, and social media

The sentiment literature gives the thesis its most novel extension. Tetlock (2007) shows that media pessimism contains information for stock prices, while Baker and Wurgler (2006) and Barber and Odean (2008) connect investor sentiment and attention to investor behaviour. Da et al. (2011) introduce Google search volume as a direct measure of retail attention, and Antweiler and Frank (2004) show that internet message-board activity contains more reliable information about trading activity and volatility than about returns. This distinction matches the thesis’s expected result: sentiment and attention may matter more for the variance of silver returns than for the mean.

Modern NLP improves measurement. FinBERT is designed for financial language and outperforms generic methods on financial sentiment tasks (Araci, 2019), while RoBERTa and Twitter-specialised models improve contextual understanding for informal, rapidly changing online language (Liu et al., 2019; Loureiro et al., 2022). This motivates using FinBERT for financial news and Twitter-RoBERTa-style models for Reddit rather than a single generic dictionary. The measurement choice is therefore grounded in the literature: domain matters, and Reddit language is not the same as financial-news language.

The evidence on Reddit-style attention is mixed but highly relevant. Studies of Wall-StreetBets and GameStop show that Reddit activity can precede or accompany extreme retail-driven market episodes, but newer evidence often finds that raw activity, comment volume, and Google Trends are stronger signals than sentiment polarity alone. A focused literature search did not surface a mature, peer-reviewed silver-only equivalent; the closest evidence comes from broader equity, Reddit, and volatility studies. That absence is useful: it creates the gap this thesis fills by testing Reddit and news sentiment directly in silver, separating tone from attention, and asking whether either adds incremental value to mean or variance forecasts.

Taken together, the literature supports the thesis design. The conditional mean is the correct object for weak- and semi-strong-form efficiency tests; the drift benchmark is the correct null; nonlinear models are useful because they make the test harder to pass; directional accuracy is interesting but secondary; and volatility is expected to be more forecastable than returns. Sentiment is therefore best treated not as a guaranteed return predictor, but as an empirical ablation that may capture attention-driven volatility, especially in a market like silver where industrial demand, monetary hedging, and retail narratives can collide.

4 Data

All models are estimated on one weekly dataset, January 2015 to May 2026, on a common Friday-to-Friday (W-FRI) calendar of 591 weeks. Seven data source families are consolidated into a single feature frame (Section 4.3); the forecast target throughout is the weekly silver log-return $r_t = \log P_t - \log P_{t-1}$. Table 1 catalogues every variable—its source, native frequency, and how it enters the models—and the rest of this section explains the choices a table cannot convey.

Table 1: Variables entering the analysis, by feature group. All features are observed at the Friday- $t-1$ close and lagged one week by each model before use, except the macro block (publication-lagged) and a best-lag ablation set. “EXOG” denotes the six cross-asset returns.

Group	Variables	Source	Freq.	Entered as
Target	<code>silver_return</code>	Yahoo Finance	Daily	weekly W-FRI log-return
AR	<code>silver_lag1/2/3</code>	Yahoo Finance	Daily	weekly log-returns, 1–3 wk lags
Cross-asset (EXOG)	<code>gold_return,</code> <code>copper_return,</code> <code>usd_return, sp500_return,</code> <code>vix_return, oil_return</code>	Yahoo Finance	Daily	weekly log-returns
Ratio (GS)	<code>gs_ratio_z</code>	Yahoo Finance	Daily	gold/silver ratio, z -scored on train
FRED daily	<code>real_rates_chg,</code> <code>breakeven_chg,</code> <code>jobless_chg</code>	FRED	Daily/Wk	weekly change
Positioning (COT)	<code>cot_mm_net_pct,</code> <code>cot_comm_net_pct</code>	CFTC	Weekly	net position / open interest
Sentiment	<code>reddit_sentiment,</code> <code>news_sentiment</code>	Reddit, GDELT	Daily	weekly mean of RoBERTa / FinBERT scores
Paid news (NEWS_PAID)	<code>news_paid_sentiment_title</code>	NewsAPI.ai	Daily	weekly mean of FinBERT title scores
Technical (TECH)	<code>macd_line,</code> <code>macd_hist,</code> <code>rsi_14w,</code> <code>mom_5w,</code> <code>roc_13/26/52w,</code> <code>price_ma13w,</code> <code>donchian_52w, bb_pct_b</code>	Yahoo Finance	Weekly	directional momentum / trend indicators
Macro (MACRO)	<code>cpi, fed_funds, ind_prod,</code> <code>m2</code>	FRED	Monthly	3 publication-availability lags each
PMI	<code>china_pmi_proxy</code>	FRED	Monthly	latest published monthly level

4.1 Sources

- **Market prices** — Yahoo Finance, daily. The three metals are global benchmarks; the equity, dollar, volatility, and oil series are US-centric:
 - silver (**SI=F**): COMEX front-month futures, the global silver benchmark — the forecast target
 - gold (**GC=F**): silver’s closest monetary substitute and dominant co-mover
 - copper (**HG=F**): industrial-demand proxy (about half of silver demand is industrial)
 - US dollar index (**DX-Y.NYB**): the dollar against a basket of major currencies; metals are dollar-priced, so a stronger dollar weighs on silver
 - S&P 500 (**^GSPC**): US large-cap equities — broad risk appetite
 - VIX (**^VIX**): 30-day implied volatility from S&P 500 options — the market’s fear / risk-off gauge
 - oil (**CL=F**): WTI crude, the US benchmark — commodity-complex and inflation proxy

The six cross-asset weekly log-returns form the EXOG set (Batten et al., 2010; Zhang & Wei, 2010).

- **Rates and inflation** — FRED, US, daily:
 - 10-year TIPS real yield (**DFII10**): the inflation-adjusted return on 10-year US Treasury bonds—the “real” interest rate. Silver pays no income, so a higher real yield makes interest-bearing bonds the more attractive store of value and tends to pull the metal down.
 - 10-year breakeven inflation (**T10YIE**): the average inflation rate markets expect over the next decade (the gap between ordinary and inflation-protected 10-year Treasury yields). Silver is held as an inflation hedge, so rising expectations can support it.
 - initial jobless claims (**ICSA**): the number of new US unemployment-benefit filings each week—a timely read on the labour market and the economy’s direction, with rising claims signalling a weakening economy.

Each enters as a weekly change.

- **Macroeconomic** — FRED, US, monthly:
 - Consumer Price Index (**CPIAUCSL**): the headline measure of realised US consumer-price inflation
 - federal funds rate (**FEDFUNDS**): the Fed’s overnight policy interest rate, which drives real yields and the dollar

- industrial production (INDPRO): the Fed’s index of US factory, mining, and utility output; about half of silver demand is industrial, so stronger output lifts physical demand for the metal
- M2 money supply (M2SL): a broad measure of money in circulation (cash plus bank deposits and near-money); rapid growth signals future inflation, which tends to lift hard assets like silver

These enter under a point-in-time availability rule rather than a naive shift: a value is visible only once it would actually have been released, with three publication lags kept per series.

- **Positioning** — CFTC (US) Commitments of Traders, COMEX silver, weekly:
 - managed-money net positioning: speculative (fund) sentiment
 - commercial net positioning: hedger and producer positioning

Each as a share of open interest.

- **Sentiment** — Reddit, GDELT, NewsAPI.ai, Google Trends (English-language / worldwide):
 - Reddit posts (r/WallStreetSilver, r/Silverbugs; via Arctic Shift): retail attention and tone, scored with Twitter-RoBERTa (Loureiro et al., 2022)
 - GDELT news headlines: global media tone, scored with FinBERT (Araci, 2019)
 - NewsAPI.ai (Event Registry) paid news: reputable English-language sources, from 2015 and at least one article per week, scored with FinBERT (Araci, 2019) on the article title
 - Google Trends “silver” search interest: worldwide retail attention (volatility chapter only)

Each text source is averaged to a weekly mean. GDELT is sparse—it begins only in late 2017 and has zero-article weeks even inside the test window—so the news series is a documented limitation; the paid NewsAPI.ai feed covers the full sample and is the cleaner news source, while the volatility chapter (Section 7) relies on Reddit and Google Trends instead.

- **Business sentiment (PMI)** — FRED, monthly: a China business-confidence proxy (CHNBSCICP02STSAM), a balance-of-opinion level standing in for industrial demand from silver’s largest physical consumer. Unlike the macro block (three pre-lagged publication lags), it enters as a single point-in-time level—the most recent value released by its publication date (a 31-day reporting lag)—taken as the week’s last observation and lagged one week by each model, exactly like the

gold/silver ratio. It is kept as its own ablation group rather than folded into the US macro set.

4.2 Technical indicators

The TECH ablation group collects momentum and trend transforms of silver’s own weekly price—MACD, RSI, rate-of-change, momentum, price-to-moving-average, Donchian position, and Bollinger %B (the full list is in Table 1)—the standard chart-analysis toolkit, included to test whether such signals carry predictive content for the return. Pure volatility and dispersion indicators are not included: they inform the conditional *variance* rather than the conditional mean a return forecast targets, so they carry little information for the return. The exact formulas and parameters are in the project repository.¹

4.3 Feature frame and split

Daily-to-weekly aggregation follows fixed rules: returns and FRED changes are summed, sentiment is averaged, and levels (`gs_ratio_z`) and COT positioning take the week’s last value. Features enter raw and contemporaneous—the macro block and a best-lag ablation set excepted—and each model applies its own one-week lag, so a feature at the Friday- $t-1$ close forecasts the Friday- t return with no intra-week look-ahead. The split is chronological, on Friday boundaries: 2015–2021 train, 2022 validation, and 2023 to May 2026 as the held-out test window (Table 2), deliberately placing the 2024–2026 bull run in the test set so performance is judged across a genuine regime shift rather than an in-sample echo of it.

Table 2: Chronological train/validation/test split (W-FRI weeks).

Split	Window	Weeks	Role
Train	2015-01 – 2021-12	365	model fitting
Validation	2022	52	model & hyperparameter selection
Test	2023-01 – 2026-05	174	held-out evaluation
Total	2015 – 2026	591	

5 Methodology

5.1 Forecasting target and benchmarks

The target for the conditional-mean analysis is the weekly silver log-return r_t of Section 2.1, forecast one week ahead. Two benchmarks frame every model. The first is a

¹<https://github.com/asugar13/silver-thesis>

naïve forecast, $\hat{r}_{t+1} = r_t$, the persistence of last week’s return; it is a weak reference for a near-white-noise target and is reported only for context. The load-bearing benchmark is the *drift*, the prevailing (expanding) mean of past returns,

$$\hat{r}_{t+1}^{\text{drift}} = \frac{1}{t} \sum_{s=1}^t r_s, \quad (9)$$

which is the random-walk-with-drift forecast and is identical to ARIMA(0, 0, 0) on an expanding window. It is the correct null for a return target (Campbell & Thompson, 2008; Welch & Goyal, 2008): a model that cannot beat the drift out of sample carries no exploitable conditional-mean information, so “beating the drift” is the operational form of rejecting the efficiency nulls of Equations (2) and (5).

Models are evaluated by *walk-forward* one-step-ahead forecasting on the held-out 2023–2026 window (Section 4.3): at each week a model is fit on data through t and predicts $t + 1$, so no future information enters a forecast. Where the model permits, two windowing schemes are compared—an *expanding* window that grows with the sample, and a *rolling* 100-week window that discards older data to track regime change. Re-estimation cadence is model-specific and noted below.

5.2 Conditional-mean models

The models form a deliberate ladder of increasing flexibility—from linear own-history, through linear public-information, to nonlinear machine learning—every rung judged against the same drift floor. Each exogenous predictor enters lagged one week (Section 4.3), and the ablation *ladder* adds feature groups cumulatively to a six-asset cross-return base (EXOG): the gold/silver ratio, technical indicators, monthly macro, the China PMI proxy, daily FRED indicators, COT positioning, and Reddit, GDELT, and reputable paid-news sentiment, up to an all-features rung. The paid-news rungs mirror the GDELT-news rungs with the title-only paid score swapped in, and the PMI proxy is added to each daily-FRED rung.

ARIMA / ARIMAX. An ARIMA(p, d, q) model with order chosen by AIC on the training and validation data; the search selects ARIMA(0, 0, 0)—that is, $r_t = \mu + \varepsilon_t$ —so the best own-history model coincides with the drift and gives the cleanest weak-form test. ARIMAX adds the lagged exogenous groups as regressors, $r_t = \mu + \beta'x_{t-1} + \varepsilon_t$, and supplies the linear semi-strong test.

VAR. A vector autoregression of silver and the cross-asset returns, stacked in a vector

y_t , with lag order p chosen by AIC,

$$y_t = c + \sum_{i=1}^p A_i y_{t-i} + \varepsilon_t, \quad (10)$$

which captures linear feedback among the assets; Granger-causality and impulse-response diagnostics are reported descriptively.

MIDAS. Mixed-data-sampling regressions keep a predictor’s recent high-frequency path instead of collapsing it to a single weekly number, and learn how much each lag should matter. A high-frequency predictor’s K lags are compressed into one term through a normalised lag-weight curve,

$$m_{j,t}(\theta_j) = \sum_{k=1}^K w_j(k; \theta_j) x_{j,t-k}, \quad \sum_{k=1}^K w_j(k; \theta_j) = 1, \quad (11)$$

where the weights follow the Beta or exponential-Almon family (Ghysels et al., 2004, 2007). The weekly forecast then regresses the silver return on its own autoregressive lags and the compressed mixed-frequency blocks,

$$r_t = \alpha + \sum_{i=1}^p \phi_i r_{t-i} + \sum_j \beta_j m_{j,t}^{(d)}(\theta_j) + \sum_m \delta_m m_{m,t}^{(M)}(\eta_m) + \sum_q \gamma_q q_{q,t-1} + \varepsilon_t, \quad (12)$$

where $m^{(d)}$ are daily MIDAS blocks, $m^{(M)}$ publication-lagged monthly-macro blocks, and $q_{q,t-1}$ weekly-release controls (jobless claims, COT). The shape parameters are estimated by nonlinear least squares (the linear coefficients profiled out), with the weight family selected on the validation window. The payoff is parsimony: rather than $K = 20$ free coefficients per daily predictor, two weight parameters trace a smooth lag profile, which is what makes daily information usable in a weekly forecast without overfitting. Two variants are run—monthly macro at three lags, and a multi-frequency model adding the six cross-asset returns at 20 daily lags (Andreou et al., 2013).

Random forest and gradient boosting. Nonlinear ensembles that admit thresholds and interactions among the lagged features—a random forest (Breiman, 2001) and gradient-boosted trees, XGBoost (Chen & Guestrin, 2016). Hyperparameters are grid-searched per ablation rung by blocked time-series cross-validation on train-plus-validation, and the model is re-fit every four weeks in the walk-forward.

LSTM. A long short-term memory recurrent network (Hochreiter & Schmidhuber, 1997) that reads a fixed window of recent weeks and learns its own representation of the past; a single recurrent layer with dropout and early stopping is used, given the small sample (a few hundred training sequences). A weekly-input model and a daily-input variant (a 60-trading-day window) are estimated, the latter testing whether

daily-resolution information helps forecast the weekly return.

The classical and tree models are run under both windowing schemes; the LSTM uses a single expanding-window training pass, since a 100-week rolling window leaves too few sequences to train stably.

5.3 Conditional-variance models

The second-moment analysis replaces the return with weekly *realised volatility*. Realised variance is additive across days, so the weekly figure is the square root of the summed daily squared returns,

$$\text{RV}_t = \sqrt{\sum_{i \in \text{week } t} r_i^2}, \quad (13)$$

a model-free volatility proxy. Its predictors are the heterogeneous-autoregressive (HAR) lags—1-week, monthly (4-week mean), and quarterly (12-week mean) trailing realised volatility—optionally extended with the six cross-asset realised volatilities, Reddit attention and sentiment-intensity measures, and Google Trends search interest; the last three form the volatility chapter’s sentiment ablation, and Google Trends enters here only, not in the conditional-mean models.

GARCH. A GARCH(1,1) model (Bollerslev, 1986; Engle, 1982), in which the return is a constant mean plus a conditionally heteroscedastic shock, $\varepsilon_t = \sigma_t z_t$, whose variance follows

$$\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2. \quad (14)$$

It is fit by maximum likelihood and re-estimated walk-forward—the parametric benchmark for conditional variance.

HAR-RV and HAR-X. The heterogeneous autoregressive model of Corsi (2009) regresses next-week realised volatility on its own trailing averages over short, medium, and long horizons; HAR-X augments that base with an exogenous block z_{t-1} ,

$$\text{RV}_t = \underbrace{\beta_0 + \beta_w \text{RV}_{t-1}^{(w)} + \beta_m \text{RV}_{t-1}^{(m)} + \beta_q \text{RV}_{t-1}^{(q)}}_{\text{HAR-RV}} + \gamma' z_{t-1} + \varepsilon_t, \quad (15)$$

where $\text{RV}^{(w)}$, $\text{RV}^{(m)}$, $\text{RV}^{(q)}$ are the trailing 1-week, 4-week, and 12-week mean realised volatilities, and z_{t-1} is the added predictor block—the six cross-asset realised volatilities, Reddit attention and intensity, or Google Trends, entered as separate ablation rungs ($z_{t-1} = \emptyset$ recovers HAR-RV). Both are estimated by OLS.

Tree ensembles. Random forest and XGBoost on the HAR-plus-cross-asset feature set, isolating any nonlinear gain over the linear HAR-X.

5.4 Evaluation protocol and significance tests

All models are scored on the same 2023–2026 test window. Point-forecast accuracy is measured by RMSE and MAE—the primary metrics, since efficiency is a conditional-mean (magnitude) claim—complemented by directional accuracy (DA) and its magnitude-weighted form (WDA) for the returns, and by direction-of-change accuracy (DCA) for volatility. Effect size relative to the drift is summarised by the Campbell–Thompson out-of-sample R^2 (Campbell & Thompson, 2008).

Significance rests on three tests. The **Diebold–Mariano** test of equal predictive accuracy (Diebold & Mariano, 1995), with Newey–West variance, compares each model to the drift floor; for returns it is reported under squared-error loss (the headline) and absolute-error loss (a heavy-tail robustness check), and for volatility under the proxy-robust QLIKE loss (Patton, 2011), which squared-error DM is too low-powered to replace on heavy-tailed realised volatility. The **Pesaran–Timmermann** test (Pesaran & Timmermann, 1992) is the matching significance test for the directional (DA/WDA) reading. Finally, a generalised spectral test (Hong & Lee, 2003) checks the own-history return series for nonlinear departures from the martingale-difference null that linear autocorrelation cannot detect.

Two robustness layers accompany the headline tests: a breakdown by calendar sub-period across 2023–2026, to confirm that performance is not driven by a single regime, and a re-run of the full battery with the exceptional 2025 bull run excluded. The load-bearing verdict throughout is the Diebold–Mariano test against the drift floor; the directional results are a secondary, economically motivated lens.

6 The Conditional Mean: Forecasting Weekly Returns

6.1 Weak form: own-history models

6.2 Semi-strong form: public information

6.3 Direction versus magnitude

6.4 Robustness

7 The Conditional Variance: Forecasting Weekly Volatility

7.1 Realised volatility is forecastable

7.2 Cross-asset spillover and nonlinearity

7.3 Sentiment and attention

8 Discussion

9 Conclusion

Weekly silver returns are indistinguishable from a martingale difference under magnitude loss, satisfying weak- and semi-strong-form efficiency in the economically relevant sense; the residual directional dependence is statistically detectable but unexploitable, which efficiency theory permits; and its episodic, regime-bound character is consistent with—though not proof of—the adaptive-markets view (Lo, 2004) in which such efficiency is the equilibrium outcome rather than the assumption. One story, three layers, no tension.

Contributions. The thesis makes four contributions. First, it evaluates a full model ladder—classical, machine-learning, and deep-learning—for *both* moments of weekly silver returns under one drift benchmark and one significance battery, so that model families are compared on a level footing rather than across incommensurable setups. Second, it makes the conditional-mean / conditional-variance factorisation explicit, turning what could read as “returns failed, volatility worked” into a single coherent efficiency statement. Third, it builds a domain-appropriate sentiment pipeline—Twitter-RoBERTa for Reddit, FinBERT for financial news—and treats sentiment as a falsifiable

ablation on both moments rather than assuming it helps. Fourth, it documents a carefully qualified set of negative results on semi-strong predictability of the mean, separating magnitude from direction and showing where an apparent directional signal fails to replicate across model classes.

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